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COMPUTATIONAL MODELLING OF GIROMILL WIND TURBINES

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ABSTRACT

Vertical axis wind turbines with straight blades, known as Giromill or H-Darrieus type, are of present interest for small-power scale applications. Their aerodynamics is more complicated than that of horizontal axis wind turbines due to the dependence of all kinematic parameters on the blade polar position reported to wind direction.

The paper shows some aspects concerning software for Giromill wind turbine designing as well as the results developed by computational modeling for various values of wind turbine's tip speed ratio. This issue has particular aspects for the low-speed wind turbines. In the modeling algorithms is used the lifting method for a wide filed of the blade's angle of attack and for a flow with boundary layer detachment. There are used recent information concerning the finite blade span. The model uses a catalog of NACA four digit airfoils developed for a wide range of geometry, angles of attacks, rough-ness, Reynolds numbers and blade spans. The methodology is applied for small power wind turbine designing used in small households. The results achieved through the computational modeling are confronted with experimental results from the literature as well as with own testing done within an aerodynamic wind tunnel of "Politehnica" University of Timisoara.

KEYWORDS

wind turbine, computational modelling, aerodynamics, vertical axis

NOMENCLATURE

A: lifting area [m^2]
$C_{P\,arb}$: power coefficient corresponding to turbine shaft [-]
C_x: drag coefficient [-]
C_y: lifting coefficient [-]
F_y: lifting force [N]
F_x: drag force [N]
F_r: radial force [N]
H: rotor height (blade length) [m]
k_T: turbine influence coefficient [-]
l: airfoil chord length [m]
$P_{T\,arb}$: wind turbine power at turbine shaft [W]
R: turbine radius [m]
$S = 2R \cdot H$: swept area [m^{2}]
v: airflow absolute velocity (wind speed) [m/s]
w: airflow relative velocity [m/s]
u: tangential velocity (blade tip speed) [m/s]
z: number of blades [-]
α: angle between tangential and absolute velocity [-]
β: angle between tangential and relative velocity [-]
φ: polar position angle [-]
$\lambda = u / v$: tip speed ratio [-]
λ_0: rated tip speed ratio (turbine rapidity) [-]
ρ: air density [kg/m^3]

1. INTRODUCTION

In the latest years the market of wind turbines has shown certain preferences for the horizontal axis wind turbines (HAWT). Despite this trend, for small power scale there is still of interest also the wind turbines with vertical axis (VAWT) especially as Girromil type (H-Darrieus, U-shaped Darrieus, and VAWT straight bladed Darrieus). The reference literature in this field shows only briefly information in which means the aerodynamics of their blades [1 - 5]. Some recent papers re-approach some of these issues [5] rather for VAWTs for household uses [7 -9]. We have taken again the issue of these wind turbines into the frame of a research project concerning a 2.5 kW wind turbine meant for certain households aiming either to save part of the taxes spend on electrical energy bought from electrical grid, or as stand alone energy source. In this context we did some computational modeling and testing in wind tunnel. Further there are shown the main conclusions concerning this research.

2. SPECIAL ASPECTS CONCERNING VAWTS

Darries wind turbines were positively evaluated due to the fact that they do not require yaw systems and because heavy parts of their power drive (gearbox, electrical generator, etc) are placed close to the ground. The main disadvantage of the classical solution is the top bearing that is guyed by three cables and three supplementary foundations for these. Hence a lot of damages have emerged. Further, the curved shape of the blades requires special technologies. These disadvantages are overtaken by the Girromill solution (H-Darrieus). But still remain some disadvantages of aerodynamic nature more severe then the absence of the yaw system, which do the VAWTs to be feasible only for small powers. These disadvantages are hence to the unsteady interaction between the airflow and the blades. This operating regime loads the sustaining structure with oscillating forces and torque which involve some designing difficulties. This unsteady load generates difficulties in using higher elevations (higher towers), tendency which obviously is used to HAWTs having certain economical benefits. Despite all these, Giromill solution for small power scale might have advantages then HAWTs. Our researches aimed to fit such of wind turbines on the top of house roofs, offering cheap solutions for reducing the household expenses on electrical energy consume.

3. OPTIMIZATION OF GIRROMIL WIND TURBINES THROUGH COMPUTATIONAL MODELING

3.1. Aerodynamic bases of wind turbine models

The software set aims engineering applications asking thus knowledge, which are used in technical applications with an enough precision. We used in this way the lifting method [6], developed on the basis of some experimental data.

In figure1 and using its related relations can be seen the connection between the blade polar position, the inlet velocity triangle as well as the aerodynamic forces on the blade. In figure 1a. can be seen the used notes, and in figure 1b and 1c are given the measures in relative position to the tangential velocity (u), for ratio $u/v>1$ and $u/v<1$. The angle field of relative velocity (w) on which depends directly the incidence angle of blade airfoil is shown in figure 2. Angle β, in the first case extends on a limited field (β_{max}, β_{min}), and in the second case the field extends on 2π radians between $+\pi$ and $-\pi$ radians. The airfoil setting angle is very low, closed to zero.

An important remark emerges: the incidence angle variation and the aerodynamic force variation depends on the polar blade position. Therefore must be used an airfoils catalog extended on the entire field of the incidence angle up to 360^0, especially when a slow wind turbine is approached.

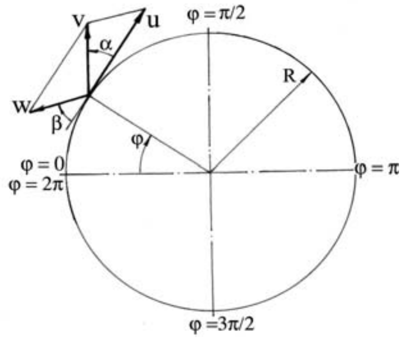

Figure 1a. Dependence of kinematic measures on the blade position

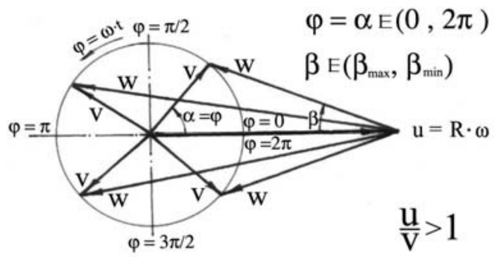

Figure 1b. Kinematics in the case of "u > v"

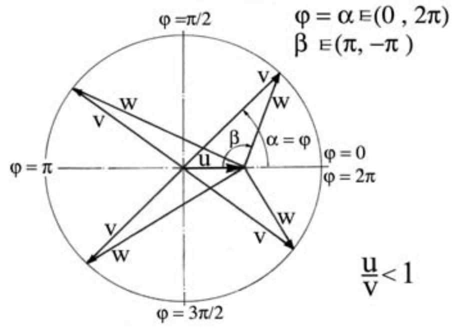

Figure 1c. Kinematics in the case of "u < v"

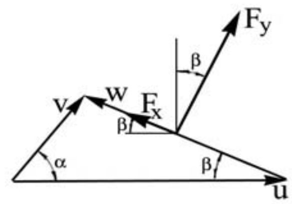

Figure 2. Velocity triangle and the directions of aerodynamic forces

The relations for the inlet velocity triangle are:

$$\alpha = \varphi \ ; \quad w^2 = u^2 + v^2 - 2 \cdot u \cdot v \cdot \cos\alpha \qquad (1)$$

$$\sin\beta = \frac{v}{w} \cdot \sin\alpha \qquad (2)$$

$$F_y = C_y \cdot \rho \cdot \frac{w^2}{2} \cdot A \qquad (3)$$

$$F_x = C_x \cdot \rho \cdot \frac{w^2}{2} \cdot A \qquad (4)$$

3.2. Airfoils catalog

We used original software developed for NACA 4 digit series airfoils. Instead the current information used for low incidence angles field, there was extended this field for the entire possible values of incidence angles (0, 360^0).

The software allows to compute the lifting, drag and moment coefficients for finite blade span, depending on: relative chamber, chamber position, relative thickness, airfoil surface roughness (smooth and standard roughness), Reynolds number. Details about the airfoils catalog can be find in reference [6].

3.1. Finite blade span effect

The Giromill wind turbine design requires corrections under the data given in the airfoils catalog (infinite span) for the case of a finite span blade. For a wide range of incidence angles we used distinctly relations for low incidences (fully attached regime), and for the detached flow in conditions of high incidences (post-stall regime). For the first case we used the Prandtl's relations, and for the second one the relations given by Viterna and Corrigen which were also used by Lissaman [10, 11].

3.2. Wake effect

Wilson's paper [4] as well as Paraschivoiu's papers [5] approach the issue of wake effect. For VAWTs the conditions downwind the turbine are much complex then those of HAWTs. The optimal solutions are usually conditioned by the wind speed field and pressure field downwind the rotor. Wilson formulates the necessity of some future researches aiming to draw nearer the theoretical and the experimental performances. The data given in the reference literature aren't enough certain for the engineering practice.

The new proposed models which considerate the pressure fall effect inside the near-wake [6] offer a new vision about the limits of extracted energy from airflow. In our software we introduces, as a temporally solution, a decreasing coefficient of wind speed through that we estimated the energetic influence of the wake conditions. For verifying the computational results we used a statistical model, which reflects a synthesis of an engineering experience, gathered in this field [12].

3.3. Some results of computational modeling

The software set achieved for Giromill wind turbines offers several possibilities for analyses developed during the wind turbine designing task. These analyses can be group in two directions:

- The influence of wind turbine rapidity on dimensionless curves, and the setting of main parameters for operational curve (rotational speeds, main dimensional features)
- The setting of geometrical details of wind turbine blades suitable with a desired operational curve, and the calculation of aerodynamic forces used for mechanical strength calculation.

One of the less certain point of the software set is the evaluation of wake effect on the velocity triangle. The software identifies certainly the inlet velocity triangle but less certainly the outlet velocity triangle as well as the dissipations inside the near-wake. These influences are evaluated through an empiric constant (k_T) found as result of some measurements done on real turbines.

Selected from the data volume obtained during the designing of the 2.5 kW wind turbine, we show further some significant examples.

Figure 3 shows the relation between the dimensionless curves and the operational curves and figure 4 gives the variation of aerodynamic radial force depending on blade polar position.

Table 1 and 2 contain the kinematic measures and aerodynamic forces computed for an operational regime.

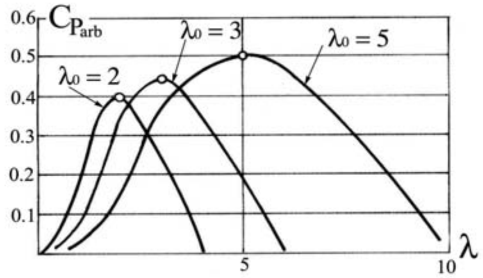

Figure 3a. Dimensionless curves for different rapidity (λ_0)

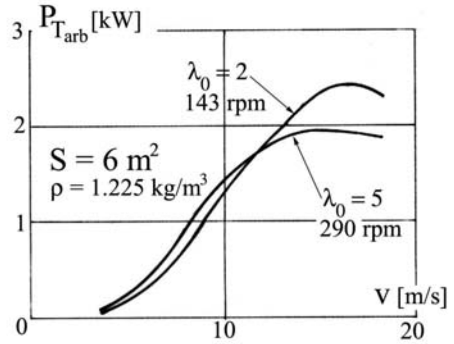

Figure 3b. Operational curves

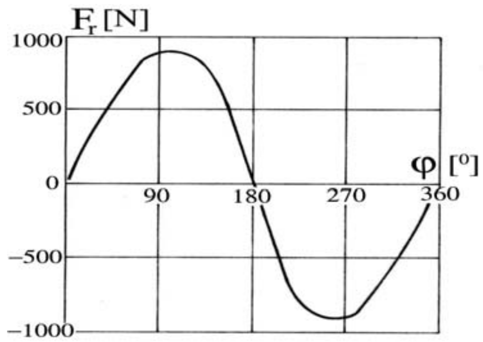

Figure 4. Radial aerodynamic force dependence on blade polar position

4. TESTING IN WIND TUNNEL

The control and the corrections on the designing algorithm of software set were done using two control methods: comparations with other technical achievemets in VAWTs filed, and testing in wind tunell upon a wind turbine model.

Table 1. Kinematic measures

Turbine T23 ; D = 2.0 m ; H = 3 m							
u = 15m/s ; v = 12 m/s; ρ = 1.225 kg/m^3							
l = 0.4 m; z = 3 ; k_T = 0.5 ; smooth surface							
φ [^{0}]	v_{1r} [m/s]	v_{1t} [m/s]	α_1 [^{0}]	β_1 [^{0}]	w_1 [m/s]	w_{1r} [m/s]	w_{1t} [m/s]
0	0	6	0	0	9	0	9
30	3	5.2	30	17.0	10.25	3	9.8
60	5.2	3	60	23.4	13.08	5.2	12
90	6	0	90	21.8	16.16	6.0	15
120	5.2	-3	120	16.1	18.73	5.2	18
150	3	-5.2	150	8.45	20,42	3	20.2
180	0	-6	180	0	21	0	21
210	-3	-5.2	-150	-8.45	20.42	-3	20.2
240	-5.2	-3	-120	-16.1	18.73	-5.2	18
270	-6	0	-90	-21.8	16.16	-6	15
300	-5.2	3	-60	-23.4	13.08	-5.2	12
330	-3	5.2	-30	-17.0	10.25	-3	9.8
360	0	6	0	0	9	0	

v_{1r}: radial component of absolute inlet velocity
v_{1t}: tangential component of absolute inlet velocity
w_{1r}: radial component of relative inlet velocity
v_{1t}: tangential component of relative inlet velocity
1: index of the inlet section

Table 2. Aerodynamic forces

φ [^{0}]	inc. [^{0}]	F_y [N]	F_{yt} [N]	F_{yr} [N]	F_x [N]	F_{xt} [N]	F_{xr} [N]
0	0	0	0	0	0.57	-0.57	0
30	17	90	27.47	86.5	1.27	-1.22	0.37
60	23.4	127	50.53	116.7	13.83	-12.7	5.5
90	21.8	200	74.12	185.3	16.15	-15	6
120	16.1	312	86.42	299.4	1.34	-1.29	0.37
150	8.4	196	28.84	194.1	8.93	-8.84	1.31
180	0	0	0	0	2.81	-2.81	0
210	-8.4	-196	28.84	-194.1	8.93	-8.84	-1.31
240	-16.1	-312	86.42	-299.4	1.34	-1.29	-0.37
270	-21.8	-200	74.12	-185.3	16.15	-15	-6
300	-23.4	-127	50.53	-116.7	13.83	-12.7	-5.5
330	-17	-90	26.47	-86.5	1.27	-1.22	-0.37
360	0	0	0	0	0.57	-0.57	0
turbine power [W]			z =1		P_{Tarb} =	584	
			z = 2		P_{Tarb} =	1128	
			z = 3		P_{Tarb} =	1692	

inc. : incidence angle
F_{yt}: tangential component of lifting force
F_{yr}: radial component of lifting force
F_{xt}: tangential component of drag force
F_{xr}: radial component of drag force

The testing was done in one of the laboratory for aerodynamics from "Politehnica" University of Timisoara, using a small Giromill rotor having the following features: diameter: 300 mm; height (balde span): 300 mm; airfiol: NACA 0020; chord length: 100 mm; number shaft): 130 W; rated tip speed ratio: 2; rated wind speed: of baldes: 3; rated power (at turbine's 20 m / s, maximum rotational speed: 2500 rpm.

Figure 5 shows this model during the testing done in wind tunnel.

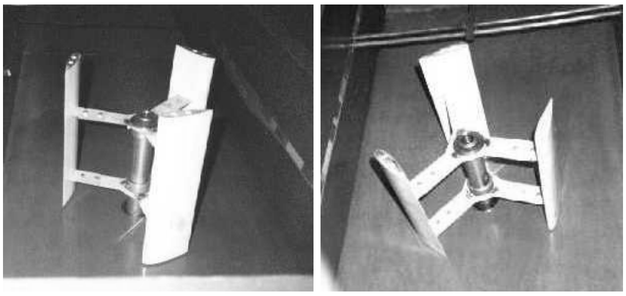

Figure 5. Small-scale model of a Giromill wind rotor tested in wind tunnel

The testing results allowed us to adjust our designing software set meant for Giromill wind turbines, enabling also to assess the constant referring to the influence of wind turbine upon the wind speed field. At the same time during testing was confirmed a variation of aerodynamic loads upon the turbine structure.

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